Appendix 3. I/O Registers

This appendix describes I/O register address and access cycles by function.

3.1 Peripheral Base Addresses

This section provides the base addresses for peripherals described in this manual. Table A3.1 shows the name, description, and the base address of each peripheral.

Table A3.1 Peripheral base address (1 of 4)

Description	Name of secure registers	Base address of secure registers in secure alias region	Name of non-secure registers	Base address of non-secure registers in non-secure alias region
Renesas Memory Protection Unit	RMPU	0x4000_0000	RMPU_NS	0x5000_0000
SRAM Control	SRAM	0x4000_2000	SRAM_NS	0x5000_2000
BUS Control	BUS	0x4000_3000	BUS_NS	0x5000_3000
Common Interrupt Controller	ICU_COMMON	0x4000_6000	ICU_COMMON_NS	0x5000_6000
CPU System Security Control Unit	CPSCU	0x4000_8000	CPSCU_NS	0x5000_8000
Direct Memory Access Controller 00	DMAC00	0x4000_A000	DMAC00_NS	0x5000_A000
Direct Memory Access Controller 01	DMAC01	0x4000_A040	DMAC01_NS	0x5000_A040
Direct Memory Access Controller 02	DMAC02	0x4000_A080	DMAC02_NS	0x5000_A080
Direct Memory Access Controller 03	DMAC03	0x4000_A0C0	DMAC03_NS	0x5000_A0C0
Direct Memory Access Controller 04	DMAC04	0x4000_A100	DMAC04_NS	0x5000_A100
Direct Memory Access Controller 05	DMAC05	0x4000_A140	DMAC05_NS	0x5000_A140
Direct Memory Access Controller 06	DMAC06	0x4000_A180	DMAC06_NS	0x5000_A180
Direct Memory Access Controller 07	DMAC07	0x4000_A1C0	DMAC07_NS	0x5000_A1C0
DMAC Module Activation 0	DMA0	0x4000_A800	DMA0_NS	0x5000_A800
Data Transfer Controller 0	DTC0	0x4000_AC00	DTC0_NS	0x5000_AC00
Interrupt Controller	ICU	0x4000_C000	ICU_NS	0x5000_C000
CPU Control Registers	CPU_CTRL	0x4000_F000	CPU_CTRL_NS	0x5000_F000
On-Chip Debug	CPU_OCD	0x4001_1000	CPU_OCD_NS	0x5001_1000
Debug Function	CPU_DBG	0x4001_B000	CPU_DBG_NS	0x5001_B000
CACHE	CACHE	0x4001_C000	CACHE_NS	0x5001_C000
TCM	TCM	0x4001_C800	TCM_NS	0x5001_C800
System Control	SYSC	0x4001_E000	SYSC_NS	0x5001_E000
Inter-Processor Communication	IPC	0x4002_0000	IPC_NS	0x5002_0000
Temperature Sensor Data	TSD	0x02C1_EDA0	TSD_NS	0x12C1_EDA0
MRAM System Register area	MRAM	0x4013_C000	MRAM_NS	0x5013_C000
Event Link Controller	ELC	0x4020_1000	ELC_NS	0x5020_1000
Realtime Clock	RTC	0x4020_2000	RTC_NS	0x5020_2000
Independent Watchdog Timer	IWDT	0x4020_2200	IWDT_NS	0x5020_2200
Clock Frequency Accuracy Measurement Circuit	CAC	0x4020_2400	CAC_NS	0x5020_2400
Watchdog Timer 0	WDT0	0x4020_2600	WDT0_NS	0x5020_2600
Watchdog Timer 1	WDT1	0x4020_2700	WDT1_NS	0x5020_2700
Module Stop Control A, B, C, D, E	MSTP	0x4020_3000	MSTP_NS	0x5020_3000

Table A3.1 Peripheral base address (2 of 4)

Description	Name of secure registers	Base address of secure registers in secure alias region	Name of non-secure registers	Base address of non-secure registers in non-secure alias region
Peripheral Security Control Unit	PSCU	0x4020_4000	PSCU_NS	0x5020_4000
Port Output Enable Module for GPT	POEG	0x4021_2000	POEG_NS	0x5021_2000
Ultra-Low Power Timer 0	ULPT0	0x4022_0000	ULPT0_NS	0x5022_0000
Ultra-Low Power Timer 1	ULPT1	0x4022_0100	ULPT1_NS	0x5022_0100
Low Power Asynchronous General Purpose Timer 0	AGT0	0x4022_1000	AGT0_NS	0x5022_1000
Low Power Asynchronous General Purpose Timer 1	AGT1	0x4022_1100	AGT1_NS	0x5022_1100
12-bit D/A Converter 0	DAC120	0x4023_3000	DAC120_NS	0x5023_3000
12-bit D/A Converter 1	DAC121	0x4023_3100	DAC121_NS	0x5023_3100
Temperature Sensor	TSN	0x4023_5000	TSN_NS	0x5023_5000
High-Speed Analog Comparator 0	ACMPHS0	0x4023_6000	ACMPHS0_NS	0x5023_6000
High-Speed Analog Comparator 1	ACMPHS1	0x4023_6100	ACMPHS1_NS	0x5023_6100
High-Speed Analog Comparator 2	ACMPHS2	0x4023_6200	ACMPHS2_NS	0x5023_6200
High-Speed Analog Comparator 3	ACMPHS3	0x4023_6300	ACMPHS3_NS	0x5023_6300
USB 2.0 FS Module	USBFS	0x4025_0000	USBFS_NS	0x5025_0000
SD Host Interface 0	SDHI0	0x4025_2000	SDHI0_NS	0x5025_2000
SD Host Interface 1	SDHI1	0x4025_2400	SDHI1_NS	0x5025_2400
Inter-Integrated Circuit 0	IIC0	0x4025_E000	IIC0_NS	0x5025_E000
Inter-Integrated Circuit 0 Wake-up Unit	IIC0WU	0x4025_E014	IIC0WU_NS	0x5025_E014
Inter-Integrated Circuit 1	IIC1	0x4025_E100	IIC1_NS	0x5025_E100
Inter-Integrated Circuit 2	IIC2	0x4025_E200	IIC2_NS	0x5025_E200
Octal Serial Peripheral Interface 0	OSPI0_B	0x4026_8000	OSPI0_B_NS	0x5026_8000
Octal Serial Peripheral Interface 1	OSPI1_B	0x4026_8400	OSPI1_B_NS	0x5026_8400
Decryption On-The-Fly 0	DOTF0	0x4026_8800	DOTF0_NS	0x5026_8800
Decryption On-The-Fly 1	DOTF1	0x4026_8900	DOTF1_NS	0x5026_8900
CRC Calculator	CRC	0x4031_0000	CRC_NS	0x5031_0000
Data Operation Circuit	DOC_B	0x4031_1000	DOC_B_NS	0x5031_1000
General PWM 32-bit Timer 0	GPT320	0x4032_2000	GPT320_NS	0x5032_2000
General PWM 32-bit Timer 1	GPT321	0x4032_2100	GPT321_NS	0x5032_2100
General PWM 32-bit Timer 2	GPT322	0x4032_2200	GPT322_NS	0x5032_2200
General PWM 32-bit Timer 3	GPT323	0x4032_2300	GPT323_NS	0x5032_2300
General PWM 32-bit Timer 4	GPT324	0x4032_2400	GPT324_NS	0x5032_2400
General PWM 32-bit Timer 5	GPT325	0x4032_2500	GPT325_NS	0x5032_2500
General PWM 32-bit Timer 6	GPT326	0x4032_2600	GPT326_NS	0x5032_2600
General PWM 32-bit Timer 7	GPT327	0x4032_2700	GPT327_NS	0x5032_2700
General PWM 32-bit Timer 8	GPT328	0x4032_2800	GPT328_NS	0x5032_2800
General PWM 32-bit Timer 9	GPT329	0x4032_2900	GPT329_NS	0x5032_2900
General PWM 32-bit Timer 10	GPT3210	0x4032_2A00	GPT3210_NS	0x5032_2A00

Table A3.1 Peripheral base address (3 of 4)

Description	Name of secure registers	Base address of secure registers in secure alias region	Name of non-secure registers	Base address of non-secure registers in non-secure alias region
General PWM 32-bit Timer 11	GPT3211	0x4032_2B00	GPT3211_NS	0x5032_2B00
General PWM 32-bit Timer 12	GPT3212	0x4032_2C00	GPT3212_NS	0x5032_2C00
General PWM 32-bit Timer 13	GPT3213	0x4032_2D00	GPT3213_NS	0x5032_2D00
Output Phase Switching Controller	GPT_OPS	0x4032_3F00	GPT_OPS_NS	0x5032_3F00
General PWM Timer Clock Controller	GPT_GTCLK	0x4032_3F10	GPT_GTCLK_NS	0x5032_3F10
PWM Delay Generation Circuit	PDG	0x4032_4000	PDG_NS	0x5032_4000
Delta-Sigma Modulators 0	DSMIF0	0x4032_A000	DSMIF0_NS	0x5032_A000
Delta-Sigma Modulators 1	DSMIF1	0x4032_A800	DSMIF1_NS	0x5032_A800
16-bit A/D Converter	ADC_B	0x4033_8000	ADC_B_NS	0x5033_8000
Serial Communication Interface 0	SCI0_B	0x4035_8000	SCI0_B_NS	0x5035_8000
Serial Communication Interface 1	SCI1_B	0x4035_8100	SCI1_B_NS	0x5035_8100
Serial Communication Interface 2	SCI2_B	0x4035_8200	SCI2_B_NS	0x5035_8200
Serial Communication Interface 3	SCI3_B	0x4035_8300	SCI3_B_NS	0x5035_8300
Serial Communication Interface 4	SCI4_B	0x4035_8400	SCI4_B_NS	0x5035_8400
Serial Communication Interface 5	SCI5_B	0x4035_8500	SCI5_B_NS	0x5035_8500
Serial Communication Interface 6	SCI6_B	0x4035_8600	SCI6_B_NS	0x5035_8600
Serial Communication Interface 7	SCI7_B	0x4035_8700	SCI7_B_NS	0x5035_8700
Serial Communication Interface 8	SCI8_B	0x4035_8800	SCI8_B_NS	0x5035_8800
Serial Communication Interface 9	SCI9_B	0x4035_8900	SCI9_B_NS	0x5035_8900
Serial Peripheral Interface 0	SPI0_B	0x4035_C000	SPI0_B_NS	0x5035_C000
Serial Peripheral Interface 1	SPI1_B	0x4035_C100	SPI1_B_NS	0x5035_C100
I3C Bus Interface	I3C	0x4035_F000	I3C_NS	0x5035_F000
Error correction circuit for MBRAM0	ECCMB0	0x4036_F200	ECCMB0_NS	0x5036_F200
Error correction circuit for MBRAM1	ECCMB1	0x4036_F300	ECCMB1_NS	0x5036_F300
CANFD Module 0	CANFD0	0x4038_0000	CANFD0_NS	0x5038_0000
CANFD Module 1	CANFD1	0x4038_2000	CANFD1_NS	0x5038_2000
EtherCAT Slave Controller	ESC	0x403A_0000	ESC_NS	0x503A_0000
Initial Configuration 1 for EtherCAT Slave Controller	ESC_INI	0x403A_4000	ESC_INI_NS	0x503A_4000
Ethernet Message Forwarding Engine	MFWD	0x403C_0000	MFWD_NS	0x503C_0000
Layer 3 Ethernet Switch Module	ESWM	0x403C_8000	ESWM_NS	0x503C_8000
Ethernet Common Agent	COMA	0x403C_9000	COMA_NS	0x503C_9000
Ethernet Agent 0	ETHA0	0x403C_A000	ETHA0_NS	0x503C_A000
Ethernet MAC 0	RMAC0	0x403C_B000	RMAC0_NS	0x503C_B000
Ethernet Agent 1	ETHA1	0x403C_C000	ETHA1_NS	0x503C_C000
Ethernet MAC 1	RMAC1	0x403C_D000	RMAC1_NS	0x503C_D000
Ethernet CPU Agent	GWCA0	0x403C_E000	GWCA0_NS	0x503C_E000
Ethernet Generic PTP Timer	GPTP	0x403E_0000	GPTP_NS	0x503E_0000
Port 0 Control Registers	PORT0	0x4040_0000	PORT0_NS	0x5040_0000
Port 1 Control Registers	PORT1	0x4040_0020	PORT1_NS	0x5040_0020

Table A3.1 Peripheral base address (4 of 4)

Description	Name of secure registers	Base address of secure registers in secure alias region	Name of non-secure registers	Base address of non-secure registers in non-secure alias region
Port 2 Control Registers	PORT2	0x4040_0040	PORT2_NS	0x5040_0040
Port 3 Control Registers	PORT3	0x4040_0060	PORT3_NS	0x5040_0060
Port 4 Control Registers	PORT4	0x4040_0080	PORT4_NS	0x5040_0080
Port 5 Control Registers	PORT5	0x4040_00A0	PORT5_NS	0x5040_00A0
Port 6 Control Registers	PORT6	0x4040_00C0	PORT6_NS	0x5040_00C0
Port 7 Control Registers	PORT7	0x4040_00E0	PORT7_NS	0x5040_00E0
Port 8 Control Registers	PORT8	0x4040_0100	PORT8_NS	0x5040_0100
Port 9 Control Registers	PORT9	0x4040_0120	PORT9_NS	0x5040_0120
Port A Control Registers	PORTA	0x4040_0140	PORTA_NS	0x5040_0140
Port B Control Registers	PORTB	0x4040_0160	PORTB_NS	0x5040_0160
Port C Control Registers	PORTC	0x4040_0180	PORTC_NS	0x5040_0180
Port D Control Registers	PORTD	0x4040_01A0	PORTD_NS	0x5040_01A0
Port E Control Registers	PORTE	0x4040_01C0	PORTE_NS	0x5040_01C0
Port F Control Registers	PORTF	0x4040_01E0	PORTF_NS	0x5040_01E0
Port G Control Registers	PORTG	0x4040_0200	PORTG_NS	0x5040_0200
Pmn Pin Function Control Register	PFS	0x4040_0800	PFS_NS	0x5040_0800

Note: Name = Peripheral name
Description = Peripheral functionality
Base address = Lowest reserved address or address used by the peripheral

3.2 Access Cycles

This section provides access cycle information for the I/O registers described in this manual.

- Registers are grouped by associated module.
- The number of access cycles indicates the number of cycles based on the specified reference clock.
- In the internal I/O area, reserved addresses that are not allocated to registers must not be accessed, otherwise operations cannot be guaranteed.
- The number of I/O access cycles depends on bus cycles of the internal peripheral bus, divided clock synchronization cycles, and wait cycles of each module. Divided clock synchronization cycles differ depending on the frequency ratio between ICLK and PCLK.
- When the frequency of ICLK is equal to that of PCLK, the number of divided clock synchronization cycles is always constant.
- When the frequency of ICLK is greater than that of PCLK, at least 1 PCLK cycle is added to the number of divided clock synchronization cycles.
- The number of write access cycles indicates the number of cycles obtained by non-bufferable write access.

Note: This applies to the number of cycles when access from the CPU does not conflict with the instruction fetching to the external memory or bus access from other bus masters such as DTC or DMAC.

Table A3.2 Access cycles (1 of 3)

Peripheral base address symbol	Address*1		Number of access cycles					
			ICLK = PCLK		ICLK > PCLK*2		Cycle unit	Related function
	From	To	Read	Write	Read	Write		
RMPU, SRAM, BUS, ICU_COMMON, CPSCU, DMAC0n, DMA0, DTC0, ICU, CPU_CTRL, CPU_OCD, CPU_DBG	0x4000_0000	0x4001_BFFF	3	2	3	2	ICLK	Renesas Memory Protection Unit, SRAM Control, BUS Control, Common Interrupt Controller, CPU System Security Control Unit, Direct memory access controller 0 n, DMAC Module Activation 0, Data Transfer Controller 0, Interrupt Controller, CPU Control Registers, On-Chip Debug, Debug Function
CACHE, TCM	0x4001_C000	0x4001_CFFF	5	4	5	4	ICLK	CM33 Cache, CM33 Tightly Coupled Memory
SYSC	0x4001_E000	0x4001_E9FF	4	3	2 to 4	1 to 3	PCLKB	System Control
SYSC	0x4001_EA00	0x4001_ED7F	7	6	5 to 7	4 to 6	PCLKB	System Control
IPC	0x4002_0000	0x4002_FFFF	3	2	3	2	ICLK	Inter-Processor Communication
ELC, RTC	0x4020_1000	0x4020_21FF	4	3	2 to 4	1 to 3	PCLKB	Event Link Controller, Realtime Clock
IWDT	0x4020_2200	0x4020_22FF	4	65	2 to 4	63 to 65	PCLKB	Independent Watchdog Timer
CAC, WDTn, MSTP, PSCU, POEG	0x4020_2400	0x4021_2FFF	4	3	2 to 4	1 to 3	PCLKB	Clock Frequency Accuracy Measurement Circuit, Watchdog Timer n, Module Stop Control, Peripheral Security Control Unit, Port Output Enable Module for GPT
ULPTn	0x4022_0000	0x4022_01FF	6	65	4 to 6	63 to 65	PCLKB	Ultra-Low Power Timer n
AGTn	0x4022_1000	0x4022_11FF	6	3	4 to 6	1 to 3	PCLKB	Low Power Asynchronous General purpose Timer n
DAC12n, TSN	0x4023_3000	0x4023_5FFF	4	3	2 to 4	1 to 3	PCLKB	12-bit D/A Converter n, Temperature Sensor
ACMPHSn	0x4023_6000	0x4023_63FF	3	3	1 to 3	1 to 3	PCLKB	High-Speed Analog Comparator n
USBFS	0x4025_0000	0x4025_03FF	5	4	3 to 5	2 to 4	PCLKB	USB 2.0 FS Module
USBFS	0x4025_0400	0x4025_04FF	4	65	2 to 4	63 to 65	PCLKB	USB 2.0 FS Module
SDHIn, IICn, OSPIn, DOTFn	0x4025_2000	0x4026_89FF	4	3	2 to 4	1 to 3	PCLKB	SD Host Interface n, Inter-Integrated Circuit n, Octal Serial Peripheral Interface n, Decryption On-The-Fly n
CRC, DOC	0x4031_0000	0x4031_1FFF	4	3	2 to 4	1 to 3	PCLKA	CRC Calculator, Data Operation Circuit

Table A3.2 Access cycles (2 of 3)

Peripheral base address symbol	Address*1		Number of access cycles					
			ICLK = PCLK		ICLK > PCLK*2		Cycle unit	Related function
	From	To	Read	Write	Read	Write		
GPT32n, GPT_OPS	0x4032_2000	0x4032_3F0F	9	6	7 to 9	4 to 6	PCLKA	General PWM 32-Bit Timer n, Output Phase Switching Controller
GPT_GTCLK	0x4032_3F10	0x4032_3F1F	4	3	2 to 4	1 to 3	PCLKA	General PWM Timer Clock Control
PDG	0x4032_4000	0x4032_4FFF	3	2	1 to 3	0 to 2	PCLKA	PWM Delay Generation Circuit
DSMIFn	0x4032_A000	0x4032_AFFF	3	3	1 to 3	1 to 3	PCLKA	Delta-Sigma Modulators n
ADC_B	0x4033_8000	0x4034_7FFF	4	3	2 to 4	1 to 3	PCLKA	A/D Converter
SCIn, SPIn, I3C	0x4035_8000	0x4035_FFFF	4	3	2 to 4	1 to 3	PCLKA	Serial Communication Interface n, Serial Peripheral Interface n, I3C Bus Interface
ECCMBn	0x4036_F200	0x4036_F3FF	5	4	3 to 5	2 to 4	PCLKA	Error correction circuit for MBRAMn
CANFDn	0x4038_0000	0x4038_3FFF	4	3	2 to 4	1 to 3	PCLKA	CANFD Module n
ESWM	0x403C_0000	0x403E_FFFF	—	—	7 to 12	2 to 4	PCLKA	Layer 3 Ethernet Switch Module
ESC	0x403A_0000	0x403A_3FFF	66	60	66 to 67	59 to 60	PCLKA	EtherCAT Slave Controller 32bit access
ESC	0x403A_0000	0x403A_3FFF	52	37	51 to 52	36 to 37	PCLKA	EtherCAT Slave Controller 16bit access
ESC	0x403A_0000	0x403A_3FFF	41	30	40 to 41	29 to 30	PCLKA	EtherCAT Slave Controller 8bit access
ESC_INI	0x403A_4000	0x403A_4FFF	4	3	3 to 4	2 to 3	PCLKA	Initial Configuration 1 for EtherCAT Slave Controller
PORTn	0x4040_0000	0x4040_01FF	4	2	4	2	ICLK	Port n Control Registers
PFS	0x4040_0800	0x4040_0FFF	8	2	8	2	ICLK	Pmn Pin Function Control Register
RSIP-E50D	-	-	3 to 5	2	1 to 6	0 to 2	PCLKA	Renesas Secure IP

Table A3.2 Access cycles (3 of 3)

Peripheral base address symbol	Address*1		Number of access cycles					
			ICLK = MRPCLK		ICLK > MRPCLK*2		Cycle unit	Related function
	From	To	Read	Write	Read	Write		
MRAM	0x4013_0000	0x4013_FFFF	4	3	2 to 4	1 to 3	MRPCLK	MRAM control

Note 1. This table only shows secure address. Access cycle of the non-secure address is the same as its secure address.
Note 2. If the number of PCLK or MRPCLK cycles is non-integer (for example 1.5), the minimum value is without the decimal point, and the maximum value is rounded up to the decimal point. For example, 1.5 to 2. 5 is 1 to 3.